Name: ______________________

SURDS,
(SIMPLIFYING SURDS)

Date: ________________

LO: I can simplify surds by finding the largest square factor.

Simplifying surds

To **simplify a surd**, find the largest **perfect square factor** of the number under the root. Use (ab) = a × b.

Method: Find the largest square number that divides into the radicand: 4, 9, 16, 25, 36, 49, 64, 81, 100...

Quick examples

	12 = (4×3) = 2 3	largest square factor 4
	50 = (25×2) = 5 2	largest square factor 25
	72 = (36×2) = 6 2	largest square factor 36
	12 = 6 × 2	not simplified: use 4
	50 = 10 × 5	need a square factor

Key idea

Find the LARGEST perfect square that divides the number, then take its root outside.

$$n = (a^2 \times b) = a \quad b$$

a^2 is the largest square factor of n

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – basic simplification

Simplify 18.

Largest square factor of 18: 9

18 = (9 × 2)

= 3 2

$\boxed{3\ 2}$

9 is the largest perfect square that divides 18.

Example 2 – larger number

Simplify 200.

Largest square factor: 100

200 = (100 × 2)

= 10 2

$\boxed{10\ 2}$

100 divides into 200, and 100 = 10.

Example 3 – with coefficient

Simplify 3 32.

32 = (16 × 2) = 4 2

3 × 4 2

= 12 2

$\boxed{12\ 2}$

Simplify the surd first, then multiply by the coefficient.

Example 4 – checking if simplified

Is 5 6 fully simplified?

Check factors of 6: 1, 2, 3, 6

No perfect square factors (other than 1)

Yes, 5 6 is fully simplified

Yes, fully simplified

A surd is simplified when no perfect square divides the radicand.

YOUR TURN – PRACTICE (deliberate practice)

1. Basic simplification

Simplify each surd.

a) 8

b) 20

c) 45

d) 28

2. Larger numbers

Simplify each surd.

a) 48

b) 75

c) 98

d) 125

3. With coefficients

Simplify fully.

a) 2 12

b) 5 8

c) 3 50

d) 4 27

4. Already simplified?

State whether each surd is fully simplified. If not, simplify it.

a) 15

b) 24

c) 3 7

d) 2 18

5. Mixed simplification

Simplify each expression fully.

a) 108 **b)** 180 **c)** 2 80 **d)** 242 **e)** 3 300

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) **12 = 2 3 because 4 is the largest square factor** ☐

Correct answer: __________

Reason: _________________

b) **12 = 4 + 8** ☐

Correct answer: __________

Reason: _________________

c) **A surd is simplified when no square number divides the radicand** ☐

Correct answer: __________

Reason:

d) **50 = 25 2** ☐

Correct answer: __________

Reason: _________________

e) **(a^2 × b) = a b** ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Find the value of n such that **n = 12 3**. Verify your answer by squaring both sides.

Expression: ____________________________

Simplified: ____________________________

b) Simplify **(2&sup4; × 3^3 × 5^2)**. Express your answer in the form a b where b has no perfect square factors.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!